

The Linear-Response Hinge

Proportional response within a restricted physical regime

IB Physics – student worksheet

Shared architecture

response = system parameter × driving quantity

A straight-line relation allows one constant to characterise a component or material. The relation applies only while the system remains within its linear-response regime.

For every case: identify the driving quantity, response, proportionality constant, units of the constant, evidence for linearity, and the observable sign that the model has ceased to apply.

1. Electrical response: an ohmic resistor

At constant temperature, a resistor obeys

$V = IR.$

A student records:

I/A	0.00	0.20	0.40	0.60	0.80
V/V	0.00	1.20	2.40	3.60	4.80

1. Identify the driving quantity and the response.

2. Calculate the resistance from the data.

3. State the units of resistance from the ratio used.

4. Describe the graph of V against I and explain how it supports a linear-response model.

5. The current is increased until the resistor becomes hot. Sketch or describe one plausible departure from the original straight line.

6. Explain why “constant temperature” is part of the model rather than an incidental detail.

Completion statement

For this system, the proportionality constant is _____; it characterises _____.

2. Elastic response: a spring

For small extensions, a spring obeys

$F = kx.$

A spring extends by 3.0 cm when a force of 1.8 N is applied.

1. Identify the driving quantity and the response.

2. Calculate the spring constant k in SI units.

3. Predict the extension for a force of 3.0 N, assuming the spring remains linear.

4. Explain what the gradient of a graph of F against x represents.

5. Distinguish the *limit of proportionality* from the statement that the spring has broken.

6. State one experimental sign that Hooke’s law is no longer an adequate model.

Completion statement

The equation gives a prediction only after the value of _____ has been measured for this spring.

3. Material response: stress and strain

Within the linear elastic region,

$$\sigma = E_Y \varepsilon, \quad \sigma = \frac{F}{A}, \quad \varepsilon = \frac{\Delta L}{L_0}.$$

A wire has $L_0 = 2.0$ m and cross-sectional area 1.5×10^{-6} m². A force of 120 N produces an extension of 0.80 mm.

1. Calculate the stress.

2. Calculate the strain.

3. Calculate Young’s modulus E_Y .

4. Why is strain dimensionless while Young’s modulus has units of pressure?

5. What feature of a stress–strain graph marks the end of linear proportionality?

6. Explain why E_Y characterises the material, whereas k characterises a particular spring geometry as well as its material.

Synthesis: one mathematical form, three different physical claims

System	Driving quantity	Response	Parameter	Condition or restriction that makes the relation linear
Ohmic resistor				
Spring				
Solid material				

Final questions

1. What is genuinely shared by $V = IR$, $F = kx$, and $\sigma = E_Y \varepsilon$? _____

2. Why is a straight line through the origin evidence for proportionality rather than proof that the law applies under all conditions? _____

3. Complete: *The constant describes the system inside a stated regime; the graph shows* _____.

The Linear-Response Hinge

Teacher guide and worked answers

Purpose. Students compare three constitutive or response relations. The common mathematics is linear proportionality; the physics lies in the parameter, the controlled conditions, and the restricted regime in which the straight-line model is valid.
Core distinction. These are not interchangeable laws. Resistance, spring constant, and Young’s modulus refer to different systems and encode different physical responses.

1. Ohmic resistor – answers

- Treat I as the controlled driving quantity and V as the measured response. Reversing the graph axes is mathematically possible, but the chosen causal language should remain explicit.
- $$R = \frac{V}{I} = \frac{4.80}{0.80} = 6.0\,\Omega.$$
- $\Omega = \text{V A}^{-1}$.
- V against I is a straight line through the origin with gradient $R = 6.0\,\Omega$.
- Heating commonly changes R . For a metallic resistor, increasing temperature usually increases resistance, so the V – I graph becomes steeper at larger currents.
- The relation $V \propto I$ is stated for constant physical conditions. Temperature change alters the component parameter and therefore the model.

Teaching emphasis

A straight graph identifies a regime and a parameter. It does not grant universal validity to the relation.

2. Spring – answers

Convert 3.0 cm = 0.030 m.

- Applied force is the driving quantity; extension is the response. In the displayed form $F = kx$, force is plotted as the dependent coordinate, so discuss the experimental choice explicitly.
- $$k = \frac{F}{x} = \frac{1.8}{0.030} = 60\,\text{N m}^{-1}.$$
- $$x = \frac{F}{k} = \frac{3.0}{60} = 0.050\,\text{m} = 5.0\,\text{cm}.$$
- The gradient of F against x is k .
- Beyond the limit of proportionality, F and x cease to be proportional. The spring may still be intact and may even remain elastic over part of the subsequent region.
- Curvature in the F – x graph, a changing gradient, or failure to return to the original length indicates failure of the simple model.

Teaching emphasis

The minus sign in $F = -kx$ describes the restoring-force direction. The magnitude relation used here is $F = kx$.

3. Stress–strain – answers

- $$\sigma = \frac{120}{1.5 \times 10^{-6}} = 8.0 \times 10^7\,\text{Pa}.$$
- 0.80 mm = 8.0×10^{-4} m, so
$$\varepsilon = \frac{8.0 \times 10^{-4}}{2.0} = 4.0 \times 10^{-4}.$$
- $$E_Y = \frac{\sigma}{\varepsilon} = \frac{8.0 \times 10^7}{4.0 \times 10^{-4}} = 2.0 \times 10^{11}\,\text{Pa}.$$
- Strain is a ratio of two lengths. Since $\sigma = E_Y \varepsilon$, E_Y has the same units as stress: pascals.
- The graph ceases to be straight; its gradient is no longer constant.
- E_Y is an intrinsic material parameter under stated conditions. A spring constant also depends on shape, length, coil diameter, wire thickness, and construction.

Completed synthesis

System	Driving quantity	Response	Parameter	Linear-regime condition
Ohmic resistor	current I	potential difference V	resistance R	Constant temperature and unchanged component properties; straight V – I graph.
Spring	applied force F	extension x	spring constant k	Below the limit of proportionality; constant gradient on the force–extension graph.
Solid material	stress σ	strain ε	Young’s modulus E_Y	Linear elastic region; stress proportional to strain.

Model synthesis answers

The shared feature is a proportional response represented by a straight line through the origin and characterised by one gradient. The physical parameter and validity conditions differ. A straight line supports proportionality only over the measured range. Departure from constant gradient is evidence that the linear-response regime has ended.

Takeaway: *A linear-response law compresses the behaviour of a system into one constant, provided the conditions and range of validity are stated.*